OPERATOR'S FIELD MANUAL · 2026 · REVERSE-ENGINEERED FROM SOURCE

# The X Algorithm, Decoded

A from-first-principles explainer *and* an operating manual. How distribution actually works — found, ranked, gated — and the seven moves that grow reach. Every claim traced to the open-source code; none of it growth-hacker lore.

## THE ONE SYSTEM THAT ORGANISES THIS MANUAL

Most "algorithm guides" blend fact and folklore into one confident voice. This one sorts *every* claim into three buckets and shows its work:

**IN THE CODE**
Present in the open-source repo. Durable, structural mechanics.
→ build strategy on it

**INFERRED**
Follows from the code, but exact magnitudes are closed.
→ use it, but caveat it

**FOLKLORE**
Repeated everywhere, absent from the code.
→ cite only with a warning

**HOW TO READ THIS**

**In a hurry?** Page 2 is the whole thing — a 60-second mental model and the cheat sheet. **Want to actually understand it?** Parts 1–7 build the system from first principles. **Want to operate?** Part 8 is the tactical guide, and every move points back to the principle that justifies it.

Sources: github.com/xai-org/x-algorithm (Apache-2.0, public — confirmed live) · phoenix/README.md · twitter/the-algorithm (2023 TweepCred & Heavy-Ranker weights) · TechCrunch & Social Media Today (Jan 20 2026). Reference only — not affiliated with X or xAI.

START HERE · THE WHOLE MANUAL IN 60 SECONDS

# If you read nothing else

X shows each person ~20–30 posts from ~500M posted daily. To choose, one Grok-based model does two jobs: it **finds** a few thousand posts it thinks you'll engage with (retrieval), then **orders** them by predicting 19 things you might do with each (ranking). It scores against *your* history, not a global notion of "good." Two gates sit around it: your **account's trust** (can you be distributed at all?) and **safety filters** (is the post allowed full reach?). That's the entire system. Everything below is detail.

### DO MORE

- Be about **one thing** §3·M1
- **Incomplete-on-purpose** posts; reply back in hour 1 §4·M2
- Media that **earns the watch** M4
- Layered + DM-worthy posts M5
- Quote *with a take* > bare repost §4

### STOP DOING

- Chasing passive-like "bangers"
- "Best time" racing — **no slot-race exists**
- Provoking blocks / mutes / reports
- Stub video; hashtag stuffing; links in body
- Topic drift; bot-like patterns

### IGNORE (RETIRED)

- "Best time to post" universals
- 6-hour decay curves
- Velocity tricks ("10 in 5 min")
- SimClusters / community lift
- Thread "completion-rate" bonus · the "65" number

## Weekly scoreboard — numbers that map to real code signals

| TRACK WEEKLY | PROXIES WHICH SIGNAL | HEALTHY |
|---|---|---|
| Replies / 1k impressions | The +75 / +13.5 reply heads (your top value) | **Up** |
| Reply-back rate, first 60 min | Are you firing the +75 author-reply-back? | **> 80%** |
| Profile clicks / post | The +12 profile-click-and-engage head | **Up** |
| Media expand / watch rate | photo-expand & video-quality-view heads | **Up vs text** |
| Out-of-network reach % | Phoenix retrieval is finding new audiences | **Up** |
| Negative-feedback incidents | block / mute / report (−74 to −369) | **~0** |
| Topic coherence, last 20 posts | How cleanly the model can place you | **1–2 lanes** |

Some are proxies (X doesn't expose dwell, embeddings, or TweepCred) — but a rough scoreboard beats flying blind.

PART 1 · FIRST PRINCIPLES

# What problem is the algorithm even solving?

Start here and everything else follows. The feed is a **selection problem under extreme constraints**.

### The constraints

- ~500M posts created per day INFERRED
- A user sees ~20–30 in a session INFERRED
- The feed must assemble in **under ~2 seconds**
- It must be **personal** — different for every viewer

### The 2026 answer

Stop hand-writing rules (“boost video,” “decay after 6h”). Instead, train **one model** to predict what each person will engage with, and let it sort everything. The repo says it plainly: IN THE CODE

*“We have eliminated every single hand-engineered feature and most heuristics from the system.”*

### The one idea the whole system rests on

There is **no global “good post.”** The model scores every candidate **against the specific person looking at it** — using *their* recent behaviour. Two identical posts get different reach for different viewers. Once you internalise this, most “growth hacks” reveal themselves as nonsense: there is no universal lever, because there is no universal viewer.

Why it changed: hand-built rules are brittle and gameable (hashtag stuffing, reply pods, posting at “peak times”). A learned model collapses thousands of rules into one function and removes the seams people used to exploit — which is exactly why the old playbooks stopped working.

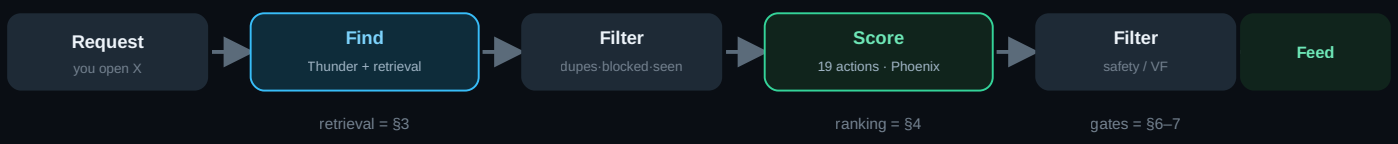
PART 2 · THE MACHINE, END TO END

# The five pieces and the path a post takes

Five components run in a fixed order. Most creators optimise the wrong one — so learn the order first.

| PIECE | ITS ONE JOB | STATUS |
|---|---|---|
| **Home Mixer** | Rust orchestrator — runs every stage in sequence and assembles the final feed. | IN CODE |
| **Thunder** | In-memory store of recent posts from accounts you follow. Supplies *in-network* candidates fast; does not rank. | IN CODE |
| **Phoenix** | The Grok-based transformer. *Finds* out-of-network posts (retrieval) and *scores* everything (ranking). The brain. | IN CODE |
| **Grox** | Reads each post as it's created — classifies category, spam, adult content, transcribes video (ASR). Feeds tags into scoring & safety. | IN CODE |
| **TweepCred** | Account-trust score (0–100). A gate on whether your posts enter wide distribution at all. | 2023 |

DIAGRAM 1 — THE LIFECYCLE OF ONE FEED REQUEST




retrieval = §3 ranking = §4 gates = §6–7

Two of these stages decide your fate: **Find** (do you even get retrieved?) and **Score** (how high do you rank?). The next two parts take each from first principles.

PART 3 · FIRST PRINCIPLES — DISCOVERY

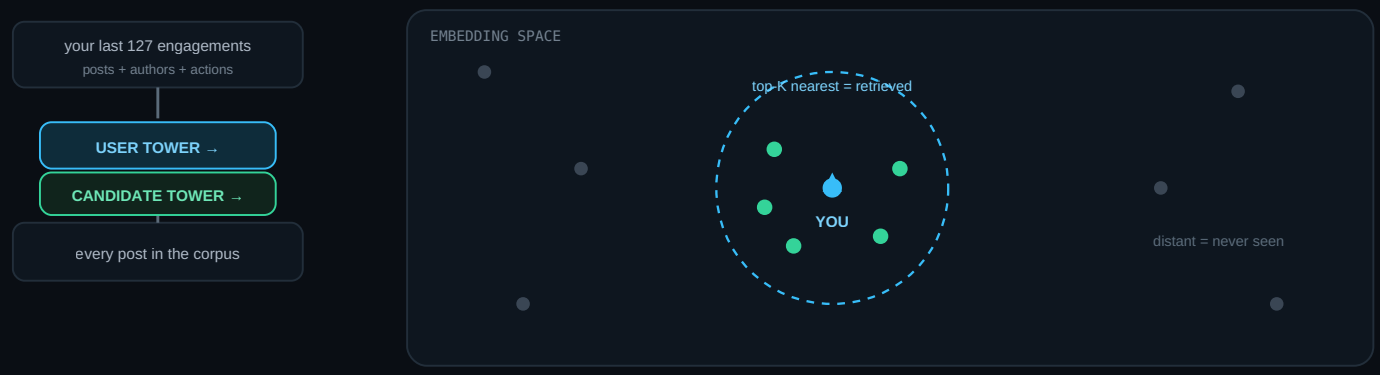
# How you get *found* (retrieval)

This is the half every other guide skips — and it's where reach to **new** audiences is actually born. It runs on one idea: **embeddings**.

## The first principle: everything becomes a point in the same space

Phoenix turns your **engagement history** into a single point — a vector — using a transformer (the "user tower"). It turns **every post** into a point too (the "candidate tower"). Both live in the same high-dimensional space. To find what to show you, it simply grabs the posts whose points are **nearest yours** (dot-product similarity), narrowing millions to about a thousand. IN THE CODE

DIAGRAM 2 — TWO-TOWER RETRIEVAL: YOU AND EVERY POST BECOME NEARBY (OR DISTANT) POINTS




Your history sets your coordinates; each post's content sets its coordinates; you're served the nearest neighbours.

### The discovery insight most creators miss

A stranger sees your post **out-of-network** when *your post's* point lands near *their* point. Your post's coordinates come from its **content** (and author). So "going wider" isn't about timing or hashtags — it's about making content whose meaning lands squarely in the region where your target audience already lives. **Legible, on-topic content is literally more discoverable**, because the model can place it. This is the first-principles reason Move 1 ("be about one thing") is move number one.

### In-network supply — Thunder

Your followers' feeds already hold your post (instant, in-memory). But Thunder only *supplies* it — Phoenix still decides where it ranks. Followers ≠ guaranteed views.

### Out-of-network supply — retrieval

New reach comes from the embedding match above. This is the growth engine: it can put you in front of people who've never heard of you — if your content is placeable.

PART 4 · FIRST PRINCIPLES — RANKING

# How you get *ranked* (scoring)

Once you're found, Phoenix predicts the probability you'll take each of **19 actions** on the post, multiplies each by a weight, and sums them. Highest score wins. IN THE CODE

$$\text{Final score} = \Sigma\ (\ \text{weight}_i \times P(\text{action}_i)\ )$$

The weights are wildly lopsided — that lopsidedness *is* the strategy. These are the 2023 published numbers; 2026 weights are learned and closed, so read the **ordering**, not the digits. 2023 REFERENCE

| ACTION | WEIGHT | ≈ VS A LIKE |
|---|---|---|
| **Reply — and you reply back** | +75.0 | **≈ 150×** |
| Direct reply | +13.5 | 27× |
| Profile opened & engaged | +12.0 | 24× |
| "Good click" (dwell 2+ min or engage) | +11.0 | 22× |
| Repost / Retweet | +1.0 | 2× |
| Like | +0.5 | baseline |
| **Mute / Block / "Show less"** | −74.0 | ≈ −148× |
| **Report** | −369.0 | catastrophic |

Bookmark and a distinct quote-tweet weight weren't in the published 2023 set (bookmark isn't one of the 2026 heads either) — we leave invented numbers out.

### Why presence beats timing — candidate isolation IN THE CODE

In the ranking transformer, candidates are masked so they **can't attend to each other** — each post is scored only against *your* history. So your post is never "out-competed" by a better one posted the same minute; **there is no slot-race.** That's the first-principles death of "best time to post" INFERRED. The first hour matters for one real reason: it's when you can fire the +75 reply-back.

### What the model knows about each viewer

Its input is your last **127 engagements**, each encoded as *post + author + action type + surface*. So it knows not just *what* you touched but *how* (a reply ≠ a like) and *where*. IN CODE

### Quote vs repost

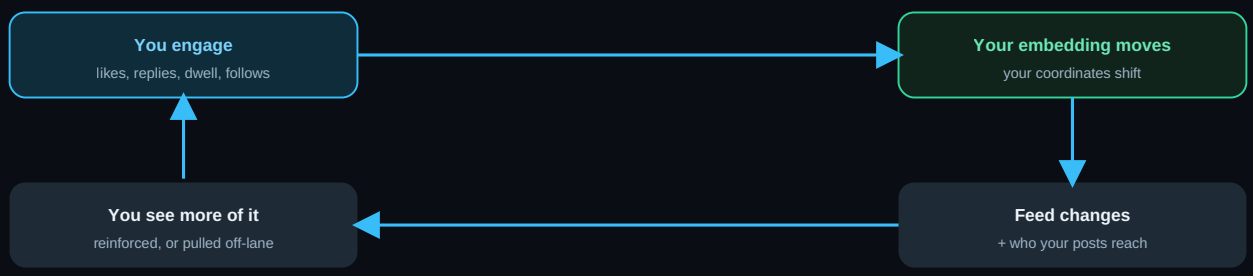
repost, quote, and quoted-click are **three separate heads** in 2026. A quote is richer — a new post that earns its own engagement *and* a click head. "Quote > repost" is sound inference, not a published number. INFERRED

PART 5 · FIRST PRINCIPLES — THE LOOP

# The reflexive loop (why your own behaviour is training data)

Retrieval and ranking both read your **engagement history**. That history is something you're writing every day — so you are continuously training the system on where to place you.

DIAGRAM 3 — THE LOOP YOU'RE ALWAYS FEEDING




OPERATIONAL CONSEQUENCE

Your **consumption** is part of your strategy. If you want to be retrieved alongside a niche, **engage genuinely inside that niche** — your embedding drifts toward the people and topics you interact with, and so does the audience your posts get matched to. Doom-scrolling off-topic content quietly relocates you. Curate your own feed like it's a positioning decision, because mechanically it is one.

This is also why bought engagement and reply-pods backfire beyond the trust hit: they teach the model a false location for you, and you get matched to people who won't actually engage — depressing the very signals that rank you.

PART 6 & 7 · THE TWO GATES AROUND THE MODEL

# The gates: trust, and safety

Scoring decides *order*. But two gates decide whether you're in the game at all — and creators routinely blame ranking for what is really a gate problem.

DIAGRAM 4 — TWO GATES, ONE SCORER




## Gate 1 — TweepCred (account trust) 2023 CODE

A PageRank-style score (0–100) on the follow/interaction graph, recalculated daily. Trust flows *in* from credible accounts that engage you. Inputs: follower quality, follow/following ratio, account age, real repeated engagement, verification, safety signals, human-vs-bot behaviour. Low trust throttles distribution **before Phoenix ever scores you**.

> FOLKLORE FLAG
>
> The famous "below 65 → only 3 posts" rule is **not** in the 2026 repo (no TweepCred at all) or the 2023 code (which only computes the score). **If a guru cites "65" as fact without caveating it, they're repeating a blog — they haven't read the code.** The direction (trust gates reach) is real; the number is folklore.

## Gate 2 — Visibility filtering & safety IN THE CODE

Separate from scoring, a `VFFilter` removes or de-amplifies posts that are deleted, spam, violent, gory, or policy-violating, and Grox classifies content for policy (PTOS) enforcement. **A post can score brilliantly and still be filtered here** — this is the engineering behind "freedom of speech, not reach." If reach collapsed despite strong engagement, suspect this gate, not your hooks.

PART 8 · WHAT CHANGED IN 2026

# New mechanics most guides haven’t caught up to

The May 2026 release added pieces that shift the game — all IN THE CODE in the update notes.

| NEW PIECE | WHAT IT IS | WHAT IT MEANS FOR YOU |
|---|---|---|
| **Promptable feeds** | Users can instruct Grok in natural language to customise their own feed. | Reach isn’t purely behaviour-driven anymore — viewers can summon or suppress topics. Being clearly *about something* makes you summonable. |
| **Ads blending** | A `home-mixer/ads` module injects ads with brand-safety tracking. | Organic posts share the feed with ads under a revenue objective — controversy that hurts brand-safety can cost reach. |
| **New candidate sources** | Phoenix MoE, topic clusters, starter packs, mutual-follow graphs. | More on-ramps to be discovered — most of them reward clear topic identity and real mutual relationships. |
| **Grox content engine** | Spam detection, category classification, ASR for video, sentiment. | Your video’s *spoken words* are now indexed; off-topic or spammy media is legible to the system. |

FIRST-PRINCIPLES READ

Every 2026 addition pushes the same way: the system understands **content meaning** better and rewards **clear identity and real relationships** over mechanical tricks. The gameable seams keep closing.

PART 9 · INTELLECTUAL HONESTY

# The limits of what anyone can know

This matters as much as the mechanics. Open-source has a hard ceiling, and most “I read the algorithm” content sails straight past it.

### What the release actually is

- The **architecture** — how data flows, the towers, the masking, the action heads. KNOWABLE
- A **frozen mini checkpoint** (128-dim, 4 layers) for demo.
- A sports demo corpus, not the live index.

### What stays closed

- The **production weights** — bigger model, *continuously* retrained on live data. CLOSED
- The learned **embeddings** (what the model thinks each post/person “means”).
- The exact action weights in 2026.

### The honest sentence to carry

You can understand the **machinery** completely and still not predict the **behaviour**, because the behaviour is a trained function nobody published. So: trust the *structure* (green), reason carefully about *magnitudes* (amber), and refuse the *folklore* (red). Anyone claiming to fully “know the X algorithm” from this code — this manual included — is overclaiming. The edge is reading the code *and saying where it goes quiet.*

PART 10 · OPERATE — THE 7 MOVES

# The operating guide

Now the payoff. Each move is wired to a principle above — the grey line names it. Run them in order; fix Move 1 before chasing Move 7.

## 1 Pick one lane and never leave it

§3 DISCOVERY

**Why (principle):** retrieval places you by embedding; a coherent history + legible content gives you stable, findable coordinates. Drift makes you unplaceable — and unfound.

**DO**

- One clear topic per post
- Keep your last ~100 posts coherent
- Engage inside your lane (you're training your own coordinates — §5)

**DON'T**

- Chase off-topic viral hits
- Run 4 unrelated themes
- Doom-scroll off-lane content

**Watch out:** one off-topic banger can relocate your embedding for weeks.

**Measure:** topic coherence of last 20 posts (1–2 lanes).

## 2 Make people reply — then reply back

§4 RANKING

**Why (principle):** "reply, and you reply back" is the single highest weight (+75, ≈150× a like). A post people *must* answer, that you answer, is the most valuable event you can manufacture.

**DO**

- Leave one open loop / invite correction
- Take a real, disagree-able stance
- Answer every reply in hour 1

**DON'T**

- Bait with fake questions / slop
- Post then vanish
- Argue to provoke blocks (M7)

**Watch out:** a passive "banger" that gets likes and no conversation is a weak post here.

**Measure:** replies / 1k impressions; reply-back rate, first 60 min.

## 3 Be in your replies for 60 minutes

§4 ISOLATION

**Why (principle):** candidate isolation = no slot-race, so "best time" is a myth. The first hour matters only because that's when replies arrive and you can fire the +75 reply-back.

**DO**

- Post when *you* can sit in replies an hour
- Treat the comment section as the post

**DON'T**

- Schedule to a "best time," then sleep
- Optimise the clock over presence

**Watch out:** auto-schedulers that post while you're offline kill Move 2.

**Measure:** % of posts you replied within 60 min.

## 4 Use media that earns the watch

§4 HEADS

**Why (principle):** photo-expand & video-quality-view are 2 of the 19 heads — text-only can't fire them. Media = extra ways to score, but only if it earns the open/watch (a stub clip fires nothing).

**DO**

- Native image/video that rewards opening
- Make the first frame earn the watch

**DON'T**

- Decorative stock stapled on
- 2-sec stub clips; YouTube links

**Watch out:** decorative media adds nothing — the head only fires on real expands/watches.

**Measure:** expand / watch rate vs your text-only baseline.

## 5 Write for dwell, and for the DM

§4 HEADS

**Why (principle):** dwell is scored twice (yes/no + milliseconds); "share via DM" is its own head, distinct from the public share — a high-intent signal.

**DO**

- Threads & layered posts that hold attention
- Write things one person sends another

**DON'T**

- Pad length without substance
- Confuse "long" with "worth dwelling on"

**Watch out:** dwell without payoff trains the opposite signal — readers bounce.

**Measure:** bookmarks + shares per post.

## 6 Look like a trusted human

§6 GATE 1

**Why (principle):** account trust gates distribution before scoring; it transfers in from credible accounts. One principle, not seven hacks.

**DO**

- Keep following ≲ followers; get Premium
- Let account age accrue
- Earn real replies from credible niche accounts

**DON'T**

- Follow/unfollow churn; buy followers
- Copy-paste / botted patterns
- Abandon an aged account

**Watch out:** the exact "65" gate is folklore (§6) — the direction is real.

**Measure:** follow ratio; negative-feedback trend.

## 7 Never hand the wrong room a reason to block you

§4 + §7

**Why (principle):** block/mute/report are explicit negative heads (−74 to −369), scored per viewer — and the safety gate can de-amplify outright. A post that delights your audience but reaches the wrong one nets negative.

**DO**

- Match the audience to the message
- Drop formats that draw mutes from outside your core

**DON'T**

- Rage-bait for impressions
- Push posts into mismatched audiences

**Watch out:** one block ≈ erases ~148 likes; one report can bury the post.

**Measure:** mutes / blocks / "not interested" per post — keep near zero.

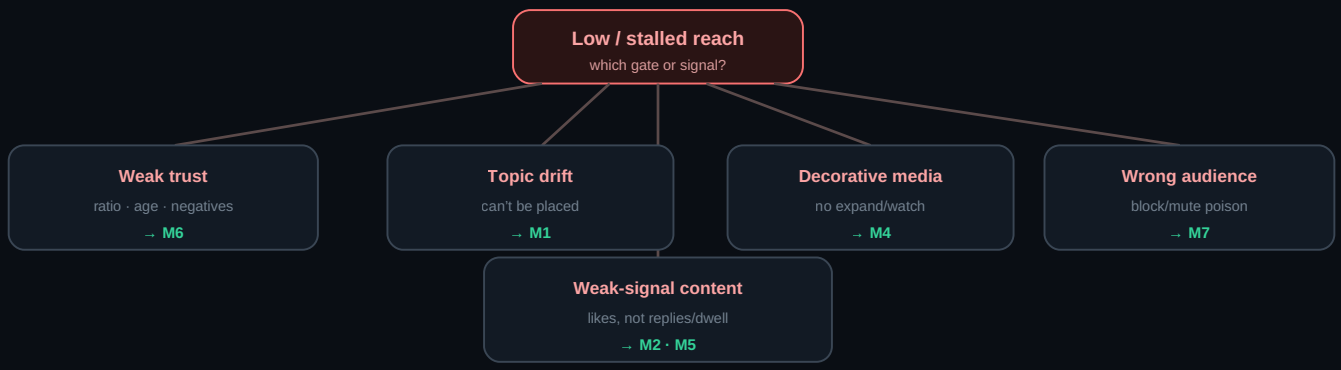
PART 10 · OPERATE — DIAGNOSE

# What’s going wrong?

Find your symptom, read the cause, run the fix. Most reach problems are one of five failure modes.

| SYMPTOM | LIKELY CAUSE | FIX |
|---|---|---|
| Good posts, low impressions | Weak trust (Gate 1), or the safety gate (Gate 2) is de-amplifying you | M6 · §7 |
| Reach stuck inside your followers | Not getting *retrieved* out-of-network — content isn’t placeable | M1 · M4 |
| Plenty of likes, reach won’t grow | Optimising the weakest signal (likes = +0.5) | M2 · M5 |
| Seen, but wrong people react badly | Audience mismatch → negative-signal poisoning | M7 · M1 |
| Threads flop | No thread bonus exists; each tweet is its own candidate | M2 · M4 |
| One viral spike, then nothing | Tolerated, not trusted — no evidence to compound | M1 · M6 |

DIAGRAM 5 — THE FAILURE MAP



PART 11 · STOP BELIEVING

# Myth vs reality

| THE MYTH | THE REALITY (FROM THE CODE) | STATUS |
|---|---|---|
| **“Post at the best time”** | Candidate isolation = no slot-race. Presence to reply back > clock. | INFERRED |
| **“10 replies in 5 min → viral”** | Engagement-velocity rules were explicitly removed in 2026. | IN CODE |
| **“Posts decay every 6 hours”** | Hand-coded recency decay gone; age is a learned embedding. | IN CODE |
| **“Hashtags expand reach”** | Hashtag matching removed — neutral at best, spammy at worst. | IN CODE |
| **“Join the right community”** | SimClusters & RealGraph removed; scoring is per-viewer. | IN CODE |
| **“Threads get a completion bonus”** | No thread bonus in code; each tweet is its own candidate. Threads help via dwell. | FOLKLORE |
| **“Below 65 TweepCred, only 3 distribute”** | Not in 2026 repo or 2023 code. Direction sound; the number is folklore. | FOLKLORE |
| **“Likes drive reach”** | Like = +0.5, the baseline. Replies (+13.5) and reply-backs (+75) do the work. | 2023 REF |

## Evidence ledger — the three buckets

**IN THE CODE — DURABLE**

Phoenix/Thunder/Home Mixer/Grox architecture · two-tower retrieval (dot-product nearest-neighbour) · 19 weighted action heads · candidate isolation · history = 127 engagements (post+author+action+surface) · media-only heads · repost/quote/quoted-click separate · each thread tweet its own candidate · VF/safety filtering · Grox category/spam/ASR · promptable feeds, ads blending (May 2026) · TweepCred 0–100 PageRank trust (2023) · hand-engineered features removed · 2023 weights +75 / +13.5 / +1 / +0.5 / −369.

**INFERRED — USE WITH CAVEATS**

Candidate isolation kills “best time” · legible content is more discoverable · media structural edge + VQV minimum-duration · quote > bare repost · 2023 weight magnitudes roughly hold in 2026 (“150× a like”) · external-link reach penalty · ~500M/day & ~1,500-candidate figures.

**FOLKLORE — DO NOT RELY**

“Below 65 → only 3 posts” · thread completion-rate bonus · exact 2026 production weights · any single universal “best time / magic hack.”

### Sources

github.com/xai-org/x-algorithm (Apache-2.0 — public, confirmed live) · phoenix/README.md · github.com/twitter/the-algorithm — ExtractTweepcred.scala & 2023 Heavy-Ranker weights · techcrunch.com (Jan 20 2026) · socialmediatoday.com · helloai.substack.com & opentweet.io (secondary).

The manual’s edge: it reads the code *and says where the code goes quiet.*